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Experiment Number : 06

Experiment Title : Demonstration of Switch Hardware and Setting up a Basic Local Area Network (LAN)

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Experiment No. 06: Demonstration of Switch Hardware and Setting up a Basic Local Area Network (LAN)

Aim

To identify the hardware interfaces of a Network Switch and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology in Cisco Packet Tracer.

Theory

What is a network switch?

A Network Switch is a networking device that connects multiple devices (computers, printers, servers, etc.) within a Local Area Network (LAN) and forwards data to the correct device using MAC addresses.

What are the main functions of a network switch?

Connects devices in a LAN

Learns MAC addresses automatically

Forwards data only to the intended device

Reduces network traffic compared to a hub

How does a switch use MAC address?

A network switch uses MAC (Media Access Control) addresses to learn where devices are connected and to forward data only to the correct device inside a LAN.

What is a MAC address table (CAM table)?

A CAM Table (Content Addressable Memory Table) is the internal table inside a network switch that stores MAC addresses and their corresponding port numbers.

Explain the plug-and-play nature of basic network switches.

The plug-and-play nature of a basic network switch refers to its ability to operate automatically without manual configuration. It learns MAC addresses, negotiates speed and duplex settings, and forwards frames intelligently as soon as devices are connected.

Hardware Components Required

- PC-PT (x3): End devices acting as network nodes. □
- Switch (2960-24TT): A 24-port Cisco Catalyst switch used to interconnect the nodes. □
- Copper Straight-Through Cables: The transmission media used for PC-to-Switch connectivity. □

Hardware and Connection Setup

- Network ID: 192.168.1.0
- Subnet Mask: 255.255.255.0

Note: PC Naming Convention:

- PC 1: Batch_Roll No. _A (e.g., S11_01_A)
- PC 2: Batch_Roll No. _B (e.g., S11_01_B)
- PC 3: Batch_Roll No. _C (e.g., S11_01_C)

Procedure

Step 1: Physical Topology Design

- Open Cisco Packet Tracer and select Network Devices > Switches > 2960. Place it in the center.
- Select End Devices > PC and place three units around the switch.
- Naming: Click on each PC label and rename them following the convention. (e.g. S11_01_A, S11_01_B, and S11_01_C).

Step 2: Hardware Identification

- Click on the Switch and select the Physical tab.
- Zoom in to observe the 24 FastEthernet ports (10/100 Mbps) and 2 GigabitEthernet ports (1000 Mbps). □
- Note the status LEDs and the Console port used for administrative access. □



Step 3: Physical Cabling

- Select the Copper Straight-Through cable (solid black line).
- Connect S11_01_A (FastEthernet 0) to Switch (FastEthernet 0/1).
- Repeat for the other two PCs using Switch ports 0/2 and 0/3.
- Observation: Notice the link lights are Orange. Click the "Fast Forward Time" button to turn them Green, indicating the Spanning Tree Protocol (STP) has cleared the ports for traffic.

Step 4: Logical Addressing (IP Configuration)

- Click S11_01_A > Desktop > IP Configuration.

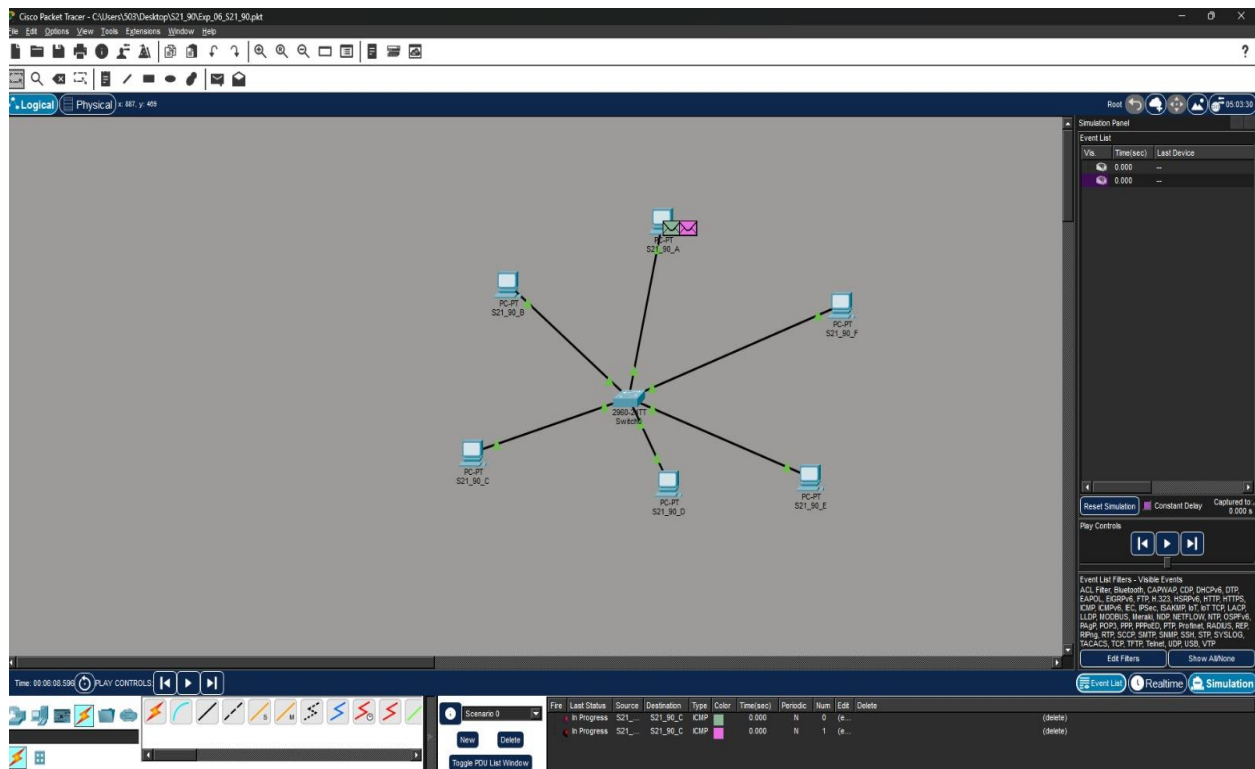
- Enter Static IP: 192.168.1.1 and Subnet: 255.255.255.0.
- Configure S11_01_B with IP 192.168.1.2 and S11_01_C with IP 192.168.1.3.

Step 5: Testing and Simulation

- **Real-time:**
- Open the Command Prompt on S11_01_A and type ping 192.168.1.3. A "Reply from..." message confirms the LAN is functional.

Simulation Mode:

- Click the Simulation tab and click Edit Filters to select only ICMP.
- Use the Add Simple PDU (Envelope) to click S11_01_A then S11_01_C.
- Click Play and observe the Switch forwarding the frame only to the destination PC.



Observations

- Media Transition: The Switch ports require time to transition from "Learning" (Orange) to "Forwarding" (Green).
- Micro-segmentation: During simulation, it was observed that the Switch does not broadcast the ICMP packet to all ports; it uses the MAC address table to send it directly to the intended recipient.
- Star Topology Logic: Disconnecting one PC did not affect the communication between the other two, proving the robustness of the Star Topology.

Conclusion

The experiment studies the hardware features of a network Switch and demonstrates the setup of a functional LAN. It shows that a Switch forwards data using MAC addresses, ensuring efficient and targeted communication.

A Switch acts as the central device in a Star Topology, improving network reliability and preventing total network failure if one connection is lost. It also enables collision-free communication by creating separate collision domains for each device.

Overall, the experiment proves that a Switch is essential for building a reliable, efficient, and high-speed LAN.